

VYOMA Innovation Challenge — Bill of Materials

MWire Labs — Suno Sutra Prototype: Khasi Education Assistant

Hardware Components

#	Component	Part / Model	Type	Qty	Est. Unit Cost (INR)	Est. Total (INR)
1	Compute module	NVIDIA Jetson Orin Nano 8GB Developer Kit	Off-the-shelf	1	28,000	28,000
2	Storage	SanDisk Ultra microSD 128GB	Off-the-shelf	1	1,200	1,200
3	Display	Waveshare 2.8" resistive touch TFT LCD	Off-the-shelf	1	2,500	2,500
4	Camera	Arducam USB camera module	Off-the-shelf	1	1,500	1,500
5	Audio output	USB speaker (generic)	Off-the-shelf	1	800	800
6	Audio input	SHOOPE USB microphone	Off-the-shelf	1	1,200	1,200
7	Touch trigger	TTP223 capacitive touch sensor module	Off-the-shelf	1	100	100
8	Real-time clock	DS3231 RTC + AT24C32 EEPROM module	Off-the-shelf	1	150	150
9	Wiring	Female-to-female 40-pin jumper wires (set)	Off-the-shelf	1	150	150
10	Networking	Ethernet cable (Cat6)	Off-the-shelf	1	250	250
11	Display cable	DisplayPort-to-HDMI adapter cable	Off-the-shelf	1	500	500
12	Power	Jetson-compatible DC power adapter/lead	Off-the-shelf	1	800	800
13	Enclosure	3D-printed outer casing with bamboo trim panel (local MSME fabrication)	Custom	1	1,500	1,500
Estimated Total (per unit)						38,650

All costs are estimated at typical Indian retail/distributor pricing as of Aug 2026 and are placeholders pending confirmed vendor invoices; part numbers for generic modules (touch sensor, RTC, jumper wires) reflect commonly available equivalents rather than a specific vendor SKU. The Jetson Orin Nano 8GB module dominates unit cost; at scale, distributor/bulk pricing and NVIDIA Inception program benefits are expected to reduce this line.

Software / Model Assets (no incremental hardware cost)

Component	Source	Type
ASR (Khasi / Hindi / English)	MWirelabs/ne-asr (own fine-tune)	Custom
Retrieval embedding model	MWirelabs/ne-embed (own, LaBSE-based)	Custom
TTS (Khasi / Garo)	MWirelabs/NE-TTS-kha-voice-01 (own, OmniVoice/Qwen3-0.6B)	Custom
KB translation (offline, build-time)	IndicTrans2 (AI4Bharat, BHASHINI-funded) + own NLLB fine-tune (Badnyal/kha-en-nllb-v0.4)	Third-party (BHASHINI-linked)
Language ID	BHASHINI Audio Language Detection (IIT-Mandi)	Third-party (BHASHINI)